Girls Build Bots — Junior Automotive Engineering

Badge-aligned workshop guide for robotics-team-led outreach

Level: Juniors (Grades 4–5) · **Badge:** Automotive Engineering — Automotive Engineering **Series:** Automotive Engineering (Design · Engineering · Manufacturing) **Format:** One 90–120 minute session · **Ratio:** 1 mentor per 4–5 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on craft supplies plus balloons the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Junior Automotive Engineering badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: from idea to road

Design it, **engineer it to perform**, manufacture it at scale. This is the test-and-measure badge: Juniors run real performance trials, protect a "passenger," and improve through fair testing.

Vocabulary: propulsion, thrust, friction, variable, fair test, data, trial, crash safety, iterate.

Automotive Engineering (90–120 minutes)

Badge theme: Testing and refining vehicles like automotive engineers — measuring performance, changing one variable at a time, and protecting the passenger.

Materials

- Balloons, flexible straws, tape, cardboard/foam chassis material, bottle-cap or CD wheels, skewer axles
- A smooth floor test track marked with tape at 1-foot intervals
- Data sheets: Trial / What we changed / Distance / Notes
- Eggs (or ping-pong balls marked with faces) as "passengers"; cotton balls, bubble wrap scraps, tissue for safety engineering
- A ramp + a soft wall (cushion/box) for the crash test

Activity 1 — Propulsion: the Balloon Engine (15 min)

Blow up a balloon and let it go — it flies from **thrust**: air pushes out backward, balloon goes forward (action–reaction). Today's engine is a balloon; the engineering is making that thrust move a car *straight and far*.

Activity 2 — Build the Balloon Car (30 min)

Teams build: chassis, free-spinning axles, wheels, and a balloon with a straw nozzle taped on. First milestone: it moves under its own power. Mentors troubleshoot the classic failures — wheels rubbing the chassis (friction!), nozzle pinched, car too heavy.

Activity 3 — Test Trials with Data (30 min)

Three timed trial rounds on the track. Each round: run, measure distance, **record it**, then change exactly **one variable** (nozzle width, chassis weight, wheel size, balloon inflation) and predict the effect before re-running. The data sheet is the point — engineers don't remember, they record. Best improvement (not best distance) gets the spotlight.

Activity 4 — Crash Safety Challenge (25 min)

New requirement: the car must carry an egg "passenger" down the ramp into the soft wall — **and the passenger must survive**. Teams engineer restraints and crumple zones from soft materials, then crash-test. Redesign once if the passenger "gets hurt." This is real automotive safety engineering: the vehicle can fail; the passenger can't.

Wrap (5 min)

Each team shares one number from their data sheet and one thing that surprised them.

> **Robot moment (optional):** Share for 60 seconds how the team runs test matches and logs data on the robot, changing one thing between tests — same trials, bigger machine.

Completion Checklist — Automotive Engineering

By the end of the session, to earn the badge, each Scout can: - [ ] Explain how thrust propels a balloon car (action–reaction) - [ ] Build a vehicle that moves under its own power - [ ] Run fair tests changing one variable at a time and record the data - [ ] Engineer passenger protection and evaluate it with a crash test

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Mentor Guide (for the robotics team)

- **You don't need a robot for this.** Balloons and cardboard carry the whole session; the "robot moment" is a 60-second add-on.
- **Enforce the data sheet.** No trial counts unless it's recorded — that's the engineering habit this badge exists to build.
- **One variable at a time.** When a team changes three things, ask "so which one worked?" and let the question land.
- **The egg has a name.** Let teams name their passenger — the safety challenge gets delightfully serious.
- **Friction is the master teacher.** Wheels rubbing, axles binding — resist fixing it for them; ask "where is it losing energy?"

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Junior Automotive Engineering badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.